

Sorry, it's my fault: Politeness, attribution, and anthropomorphism in managing generative AI hallucinations

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ABSTRACT

While generative artificial intelligence (AI) has revolutionized various fields, it also presents a significant challenge: 'hallucinations'—plausible but inaccurate information generated by AI systems. Because hallucinations are difficult to prevent entirely, it is essential for generative AI systems to address these inaccuracies effectively. This study investigates how generative AI response strategies to hallucinations affect user satisfaction and tolerance. We examined the impact of politeness (Gratitude vs. Apology) and attribution (Internal vs. External) strategies, as well as AI anthropomorphism, on user reactions. In a 2×2 online experiment with 369 ChatGPT users, participants were randomly assigned to one of four response strategy conditions. Results show that users reported the highest satisfaction when the AI apologized and accepted internal responsibility for the error. This effect was particularly pronounced among users who perceived the AI as less human-like, though positive reactions were also observed among users who anthropomorphized the AI. Moreover, user satisfaction mediated the relationship between the AI's apology/internal attribution and tolerance for hallucinations. This indirect effect was strongest among those who perceived the AI as less human-like. These findings offer theoretical insights into how social response strategies shape user tolerance of AI errors and provide practical guidance for designing more trustworthy and human-centered AI.

1. Introduction

While generative artificial intelligence (AI) excels at generating contextualized responses based on vast amounts of data, it can also exhibit "hallucination," which refers to generating inaccurate or misleading information (Pavlik, 2023; Stryker & Scapicchio, 2024). Hallucinations not only undermine the credibility of generative AI technologies (Marr, 2023) but also risk causing social disruption by leading users to accept misinformation as fact (Kabir et al., 2023; Pavlik, 2023; Stahl & Eke, 2024). Misinformation can lead to a lack of trust in generative AI; therefore, it is important for generative AI to adequately inform users that it has generated misinformation.

Recent research concentrates on the phenomenon of hallucinations in generative AI and their causes (Borji, 2023; Kabir et al., 2023; Marr, 2023; Zhang et al., 2023b). Given the difficulty of completely preventing hallucinations from a technical perspective (Athaluri et al., 2023; Ji et al., 2023), it is crucial for generative AIs to have a strategy to deal with this issue. However, the most satisfactory strategy for users remains

unclear. Despite the rapid progress in conversational generative AI powered by large language models, the issue of producing incorrect answers remains. For practitioners, a major challenge is to develop strategies that enhance user satisfaction when these AI systems make mistakes, while also encouraging users to be more tolerant of such errors.

Generative AIs can conduct natural and fluent conversations by effectively emulating the structure of human language (Brown et al., 2020). Owing to this linguistic sophistication, users often perceive generative AI as a social being (Nißen et al., 2022). Consequently, generative AI must adhere to basic social norms and politeness as humans do in social interactions. In this context, generative AI can draw on interpersonal communication strategies to respond effectively to errors and maintain positive relationships with users, particularly in situations involving hallucinations.

These strategies, which people use to choose the appropriate communication skills for a given situation can help AI consider a user's feelings, communicate clearly, and respond appropriately. Maintaining

Abbreviations: AI, Artificial intelligence; CASA, Computers Are Social Actors; CMC, Computer-mediated communication; GPT, Generative Pre-trained Transformer; HCI, Human-computer interaction; KMO, Kaiser-Meyer-Olkin; LLM, Large language model.

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a positive image of both the AI and the user during the communication process is essential for smooth interaction, and this can be achieved through the strategic use of politeness to protect and maintain the faces of both parties (Brown & Levinson, 1987).

The self-image that an individual tries to maintain in order to be judged positively by others is called face (Goffman, 1955). According to politeness theory, people have a basic need to maintain positive and negative faces in social interactions (Brown & Levinson, 1987). A positive face refers to the desire to receive positive evaluation from others. Compliments and expressions of gratitude are ways to honor another person's positive faces. People strategically use gratitude to deal with another person's dissatisfaction when something goes wrong during a conversation (Ahmadi & Fakhimi, 2021; Song et al., 2023; You et al., 2020).

A negative face refers to an individual's desire to act independently and autonomously without interference or coercion (Brown & Levinson, 1987). An apology honors another person's negative face by acknowledging the discomfort you caused them. By apologizing, the speaker acknowledges the wrongdoing and demonstrates a willingness to resolve the resulting problem, helping the other person regain autonomy. Apologies and expressions of gratitude are considered key elements of social interaction between users and AI agents in computer-mediated communication (CMC) (Lee & Lee, 2022; Li & Zhao, 2022; Pompe et al., 2022; Song et al., 2023; Xu & Howard, 2022; You et al., 2020). Moreover, trust in AI is often more fragile than trust in human agents and more difficult to restore once violated (Madhavan & Wiegmann, 2007). This fragility underscores the importance of socially appropriate communication strategies—such as apologies and clear attribution—in preserving user trust, especially in response to hallucinations.

People use attribution strategies to interpret and explain situations or causes to reduce misunderstandings or conflicts that may arise during communication. Attribution refers to the cognitive effort that an individual makes to infer the cause of a particular behavior or event (Heider, 1958). Heider (1958) divided attribution into internal and external. Internal attribution posits that an individual's behavior is determined by internal characteristics, such as personality, attitude, and abilities. In contrast, external attribution suggests that an individual's behavior is determined by external factors, such as context, environment, and luck. Users' perceptions of AI and their satisfaction with it are significantly influenced by how they interpret the AI's actions and outcomes. This interpretation process, often referred to as attribution, plays a crucial role in shaping the user experience (Huo et al., 2022; Yue & Li, 2023).

The way AI service providers explain the cause of a problem or situation can also affect users' perceptions of the service. When providers employ an external attribution strategy, they tend to perceive a lack of responsibility and accountability on the part of the service provider when they fail to deliver services (Fu et al., 2015; Srivastava & Gosain, 2020). Conversely, when an internal attribution strategy is adopted, users perceive that the firm accepts responsibility and shows a willingness to improve, leading to increased satisfaction with the firm's brand (Yuan et al., 2016). Research on how to effectively address AI hallucinations through attribution strategies is limited, leaving a significant gap in understanding how users respond to AI-generated misinformation.

The impact of politeness and attribution strategies on users can vary depending on how similar they perceive AI to be to a human. The cognitive process of attributing human characteristics to non-human objects (e.g., animals, objects, and technology) is called anthropomorphism (Epley, 2018). Through anthropomorphization, people tend to perceive non-human objects such as machines or AI as social beings (Vollmer et al., 2018). The more people anthropomorphize a non-human object and see it as similar to a human, the more they feel close to it and respond favorably to it (Chandra et al., 2022; Kim et al., 2019; Xie et al., 2023). According to Liu and Tao (2022), users who anthropomorphize AI services respond more favorably. Therefore, users who anthropomorphize generative AI are more likely to accept the generative AI's

polite attitude and clear attributions positively.

To situate interpersonal communication strategies within a cohesive theoretical framework, this study draws on two complementary perspectives: Human-Centered Artificial Intelligence (HCAI) and the Computers Are Social Actors (CASA) paradigm. HCAI underscores the importance of transparent, respectful, and user-centered responses to system failures, framing errors as opportunities for meaningful interaction rather than mere technical malfunctions (Schmager et al., 2025; Shneiderman, 2020). In parallel, CASA posits that users often apply human social norms—such as politeness and reciprocity—even to non-human agents, especially when minimal social cues are present (Nass & Moon, 2000).

Together, these frameworks provide a foundation for understanding how users interpret AI's responses to hallucinations through the lens of social expectations and perceived humanness. Based on this integrated view, this study investigates how two interpersonal strategies—politeness and attribution—shape user satisfaction and tolerance in hallucination scenarios. By further considering users' perceptions of anthropomorphism, the study aims to identify which communication styles are most effective and offer actionable insights for designing emotionally intelligent, human-centered AI systems.

2. Theoretical background

2.1. Generative AI hallucinations and human-centered communication responses

Generative AI refers to a form of artificial intelligence that creates new content—such as text, images, and audio—based on patterns learned from large-scale datasets (Pavlik, 2023). These systems are typically built on large language models (LLMs), which predict the next word in a sentence using statistical probabilities derived from vast textual corpora (Bail, 2024). OpenAI's ChatGPT, Google's Bard, and Microsoft's Bing are prominent examples of generative AI systems that produce contextually appropriate responses across various domains, including chatbots, translation, content creation, and coding (Stryker & Scapicchio, 2024).

Despite their widespread use and rapid advancement, generative AI systems are prone to producing hallucinations—outputs that sound plausible but are factually incorrect. This phenomenon has emerged as a key challenge, particularly as user adoption continues to rise. Recent studies show that even state-of-the-art models like ChatGPT still frequently generate inaccurate information (De Vynck, 2023; Kabir et al., 2023). For instance, Kabir et al. (2023) found that over half of ChatGPT's responses to factual queries contained errors. Users may also accept these hallucinations as true when they are presented in a persuasive and coherent manner, raising ethical concerns about misinformation (Stahl & Eke, 2024).

Moreover, hallucinated outputs are not always neutral in content. Large language models (LLMs) have been shown to reflect and even amplify existing societal biases, including those related to gender, race, and ethnicity (An et al., 2025). In such cases, hallucinations can do more than simply misrepresent facts—they can reinforce harmful stereotypes and contribute to social and ethical harms by reproducing inequities present in training data.

Although the root causes of hallucinations are not fully understood, scholars point to limitations in both model architecture and training data. One contributing factor is the use of AI-generated content as future training material, a process known as self-training, which can degrade model accuracy over time (Shumailov et al., 2023). Probabilistic modeling in LLMs can also lead to inconsistent or contradictory outputs (Borji, 2023; Ji et al., 2023).

Given these limitations, eliminating hallucinations remains a difficult technical challenge. Moreover, their frequent and unpredictable nature can significantly undermine user trust (Marr, 2023). In response, scholars increasingly argue that technical improvements must be

complemented by clear, user-centered communication strategies and effective expectation management frameworks (Kinney et al., 2024). Transparent explanations and dynamic interaction designs can help sustain user trust even when AI outputs are flawed (Ou et al., 2024).

Within this broader shift, the Human-Centered Artificial Intelligence (HCAI) perspective offers a compelling framework. HCAI emphasizes the design of AI systems that promote human well-being, uphold ethical standards, and support user autonomy through transparency and accountability (Schmager et al., 2025). From this perspective, hallucinations are not merely system malfunctions but critical interaction points—opportunities for AI to engage users in supportive, respectful, and socially meaningful ways (Shneiderman, 2020).

For example, when AI acknowledges its own errors and responds politely, it may help preserve user trust and improve satisfaction. Prior work shows that users increasingly expect conversational agents like ChatGPT to communicate in ways consistent with human social norms and to interact transparently to enhance credibility and acceptance (Kinney et al., 2024; Ou et al., 2024). This is especially important as LLM-based agents are often perceived as humanlike in their conversational fluency.

From a human-centered perspective, appropriate communication—such as expressing politeness and clearly attributing responsibility—can help mitigate the negative effects of hallucinations (Huo et al., 2022; Song et al., 2023). These social strategies enable AI systems not only to fulfill their functional roles but also to foster more relational and trust-based interactions with users. Building on this perspective, this study investigates how two interpersonal communication strategies—politeness and attribution—can shape user satisfaction and tolerance in hallucination scenarios.

2.2. Generative AI's strategy for dealing with hallucinations

2.2.1. Politeness strategy

Politeness is the self-image that an individual tries to maintain in social interactions to be positively evaluated by others (Goffman, 1955). People strategically use politeness when communicating to appear polite. According to politeness theory, people desire to maintain their positive and negative face (Brown & Levinson, 1987). A positive face refers to the desire to be respected and recognized by others and can be satisfied through praise or gratitude (Brown & Levinson, 1987). For example, when someone is helpful to you, you can honor their positive face by saying "thank you" or "that was really helpful." Honoring another person's positive face through expressions of gratitude makes him or her feel valued, which is beneficial for interpersonal communication (Algoe et al., 2016).

In contrast, people desire to act independently without interference or coercion from others, which is called a negative face. Speakers can honor another person's negative face by emphasizing that they can maintain their freedom and independence through apologetic expressions (Brown & Levinson, 1987; Song et al., 2023). For instance, in a situation where someone asks for a favor, the speaker can honor the other person's negative face by expressing, "I'm sorry for asking you to do that."

Apologies are effective in resolving problematic situations because they respect the other person's autonomy and make the request or favor feel less coercive (Roschk & Kaiser, 2013). Expressing gratitude to acknowledge the other person's positive face and apologizing to respect their negative face are both common politeness strategies in social interactions.

Recently, politeness strategies have been incorporated into CMC. Previous research concentrates on the impact of service providers' politeness strategies on user reactions in the context of service failure, which occurs when a service provider fails to fulfill a user request. Gratitude strategies foster positive interactions, build rapport (Algoe et al., 2016), and help sustain social relationships (Bock et al., 2021; Ma et al., 2017). According to You et al. (2020), expressions of gratitude

from service providers are more effective in improving users' self-esteem and satisfaction because expressions of gratitude that recognize patience and merit increase users' self-esteem, leading to positive responses. Expressing gratitude through the voice interface of an autonomous vehicle has also been shown to increase drivers' perception of social presence, which increases their trust in the vehicle (Lee & Lee, 2022).

Some research suggests that apologizing is more likely to increase user satisfaction than gratitude. Choi et al. (2021) demonstrated that apologies from service robots effectively enhance user service satisfaction. When a robot apologizes to compensate for a mistake, users perceive the robot as kinder and more competent. The way an apology is delivered significantly affects users' reactions. When users are dissatisfied with a service, they are less likely to complain if the service provider uses empathetic language in the apology (Roschk & Kaiser, 2013). Apologies accompanied by remorse are also effective in resolving user dissatisfaction. For instance, Pompe et al. (2022) found that when a voice-activated robot made an error, acknowledged its actions, and explained the reason for the mistake, it restored user trust more effectively than when it did not apologize. Additionally, when the robot apologizes and expresses shyness about the mistake or problem (Li & Zhao, 2022) or shows emotions such as embarrassment or sadness (Xu & Howard, 2022), user dissatisfaction with the robot is significantly reduced.

Previous studies show that in the context of CMC, expressions of gratitude and apology can play an important role in creating positive user attitudes toward a service (Lee & Lee, 2022; Pompe et al., 2022; Roschk & Kaiser, 2013; Song et al., 2023). Even in the context of hallucinations, a generative AI politeness strategy can positively influence user attitudes toward generative AI. Expressions of gratitude strengthen the bond between service providers and users (Algoe et al., 2016; Ma et al., 2017) and contribute to users' self-esteem by recognizing their perseverance and contributions (You et al., 2020).

By contrast, an apology serves as an effective response to user complaints because it conveys regret and a clear willingness to resolve the issue (Choi et al., 2021; Pompe et al., 2022; Roschk & Kaiser, 2013). The present study specifically examines a context in which users have already recognized and pointed out the AI's hallucination error. In such cases, expressions of gratitude are limited to signaling responsibility or a willingness to improve, rather than directly addressing the user's concern, and may therefore appear incongruent or even manipulative (Kim et al., 2025). Conversely, an apology better aligns with user expectations for responsible and emotionally appropriate communication, enhancing perceived authenticity and, ultimately, satisfaction. Prior research supports this mechanism, showing that AI systems that respond in socially appropriate and emotionally congruent ways are more likely to be perceived as trustworthy and authentic (Vo et al., 2025; Zhang et al., 2024). Therefore, we expect users to be more satisfied when the generative AI expresses an apology than when it expresses gratitude in the context of hallucinations.

Hypothesis 1. User satisfaction tends to be higher when generative AI expresses an apology rather than gratitude

2.2.2. Attribution strategy

Attribution refers to the cognitive process of reasoning and explaining the causes of one's own behavior or that of others (Heider, 1958). According to attribution theory, how an individual attributes the cause of others' behavior or events determines how they react to them (Heider, 1958; Weiner, 1994). When people face negative situations, they often seek to identify the cause of the outcome, attributing responsibility either to personal characteristics or external circumstances (Heider, 1958; Weiner, 1985). Heider (1958) categorized attribution into two types: internal and external. Internal attribution attributes an individual's behavior to internal characteristics such as personality, attitudes, or abilities. For instance, if someone receives a good grade on a test, they may see it as a result of their own effort or ability—this is

internal attribution. On the other hand, external attribution attributes the behavior to external factors, such as situational circumstances, the environment, or luck. For example, if a person receives a good grade on a test, they might attribute it to the ease of the questions or simply luck—this is external attribution.

Attribution plays a crucial role in shaping user responses, even in the context of human-computer interactions. Previous researchers have examined the impact of user attribution and service provider attribution on user responses. First, when users attribute the cause of an issue to themselves, rather than to AI (i.e., internal attribution), it is positively related to their acceptance of AI technology. Users with high self-responsibility tend to believe that when something goes wrong, the problem lies in their interaction with the AI, rather than in its error (Huo et al., 2022). According to Huo et al. (2022), users' internal attributes lead to higher trust in AI and have a positive impact on their acceptance of AI technology. On the other hand, when the use of AI services does not meet consumer expectations or needs, people tend to externally attribute responsibility for negative outcomes to AI (Yue & Li, 2023). According to Weitzl et al. (2018), the more users externally attribute the cause of a problem, the more dissatisfied they are. Moreover, they are more likely to tell others about their negative experiences.

In negative situations where a service is not delivered properly, the attributes of the service provider also have an important impact on the user. When service delivery fails, providers adopt attribution strategies to maintain user satisfaction. However, the level of user dissatisfaction may vary depending on attribution. In the context of explaining the causes of service failure, when service providers attribute the failure to external causes beyond their control (i.e., external attribution), it is known to have a negative impact on user satisfaction (Fu et al., 2015; Srivastava & Gosain, 2020). When service providers attribute service failures to external causes that are not their responsibility, users are likely to perceive the service provider as lacking responsibility and accountability. This perception can undermine trust and faith in the service, leading to lower customer satisfaction (Fu et al., 2015) and triggering anger and negative perceptions of the brand (Lee, 2005). Yuan et al. (2016) found that a firm's internal attribution strategy, which acknowledges its responsibility for failure, positively shapes users' brand attitudes, while an external attribution strategy negatively influences users' attitudes.

Existing research has shown that external attribution strategies can negatively impact users by creating an impression of a lack of accountability on the part of service providers (Lee, 2005; Fu et al., 2015; Srivastava & Gosain, 2020). By internally attributing the cause of the error, generative AI can acknowledge its responsibility and demonstrate a willingness to improve (Yuan et al., 2016). In other words, the internal attribution strategy gives the impression that AI is clearly aware of its limitations and indirectly conveys that it is working to improve them. Therefore, when generative AI employs an internal attribution strategy, users are likely to perceive it as more transparent and accountable than when an external attribution approach is employed. This, in turn, can lead to higher user satisfaction with the AI response even in the face of service failure.

Hypothesis 2. User satisfaction is higher when generative AI attributes hallucinations to internal factors rather than external ones

2.2.3. Strategies for mitigating hallucinations in generative AI based on politeness and attribution

Generative AIs can create new content based on training data; however, in this process, misinformation or unrealistic data are generated. To effectively respond to the problem of hallucinations, this study proposes a communication strategy for generative AI based on the politeness and attribution strategies used in interpersonal communication and human-computer interaction (Table 1). According to politeness theory, people desire to maintain their positive and negative faces during interactions (Brown & Levinson, 1987). By respecting the user's

Table 1
Response strategies to AI hallucinations.

Strategy	Internal Attribution	External Attribution
Gratitude	"Thank you for pointing out the error. It seems I misunderstood your question."	"Thank you for pointing out the error. It seems the external source I referenced (e.g., a dictionary, website, or report) contained some confusing information."
	"I apologize for the incorrect answer. It seems I misunderstood your question."	"I apologize for the incorrect answer. It seems the external source I referenced (e.g., a dictionary, website, or report) contained some confusing information."

positive and negative faces through expressing gratitude or apologizing, Generative AI can strengthen the relationship and increase user satisfaction. According to the attribution theory, attitudes and behaviors vary depending on whether an individual attributes the cause of an event to personal characteristics or external circumstances (Heider, 1958). In the context of generative AI, politeness and attribution strategies can be applied to help users understand and interpret AI output, potentially mitigating negative reactions to hallucinations or errors.

Most studies have conceptualized the phenomenon of hallucinations in generative AI (Borji, 2023; Zhang et al., 2023b) and discussed its impact on users (Kabir et al., 2023; Marr, 2023). Because large-scale language models such as ChatGPT can sometimes make incorrect inferences or present errors contained in the training data to users, it is technically difficult to completely prevent hallucinations (Ji et al., 2023). Therefore, it is crucial for generative AI to develop strategies to respond to errors when hallucinations occur. These strategies can help maintain user satisfaction when the generative AI provides incorrect information. However, there has been little discussion regarding which of these four strategies can most effectively increase user satisfaction.

Apologies directly address errors, express regret, and demonstrate willingness to resolve issues, which can lead to higher user satisfaction (Choi et al., 2021; Pompe et al., 2022; Roschk & Kaiser, 2013). Therefore, in the context of hallucinations in generative AI, apology is likely more effective than expressions of gratitude. When combined with an internal attribution strategy, apology is particularly effective. This approach implies that the company transparently acknowledges service issues and is committed to resolving them (Yuan et al., 2016). However, because internal attribution involves admitting flaws in a company's service, users may perceive the error as more serious. In such cases, an apology can help mitigate user dissatisfaction and increase tolerance for errors by formally acknowledging the seriousness of the problem and inconvenience to the customer (Song et al., 2023).

Hypothesis 3. When generative AI attributes the cause of errors internally while apologizing, user satisfaction with generative AI is relatively higher.

2.3. Moderating effect of anthropomorphism

Anthropomorphism refers to the cognitive process of attributing human characteristics to non-human objects (e.g., animals, objects, and technology) (Epley, 2018). This concept encompasses both physical and psychological human characteristics (Zhang et al., 2021), allowing people to interact with non-human objects in a more familiar manner (Chandra et al., 2022). Anthropomorphism is particularly important in human-computer interaction (HCI) as it helps users perceive technology as a social actor, thus improving their user experience (Xie et al., 2023). For example, when AI chatbots are designed with human-like characteristics, users feel more trust and connection in their interactions (Chandra et al., 2022; Xie et al., 2023). Kim et al. (2019) found that the more human-like a robot was perceived, the more positively users evaluated its emotional and cognitive aspects. These findings underscore the role of anthropomorphism in enhancing user experience and

satisfaction with AI (Araujo, 2018; Sheehan et al., 2020).

The Computers Are Social Actors (CASA) paradigm provides an important theoretical framework for human-technology interaction. According to this paradigm, people tend to treat machines as real people during interactions even when aware of their non-human nature (Reeves & Nass, 1996). This explains how non-human objects, such as AI, can be perceived as social actors. People unconsciously treat AIs as social entities (Vollmer et al., 2018), expect them to exhibit human-like characteristics. Conversational generative AIs, with human-like language structures, allow natural interactions (Brown et al., 2020). This sophistication in language use may lead users to perceive these AIs as social entities (Nißen et al., 2022), potentially influencing how they interact with and relate to them.

Anthropomorphism is closely related to attribution theory. Attribution theory explains how people interpret others' actions and infer their causes, a process that can also be applied to anthropomorphized objects (Kim & Song, 2021). When people perceive technologies, such as AI or robots, as human-like, they tend to attribute their behaviors in ways similar to how they behave with humans. For instance, when an AI makes a mistake or exhibits unexpected behavior, users seek to understand the cause of the behavior and often attempt to interpret it based on human-like intentions or emotions. Kim and Song (2021) found that in a stock investment system, AI agents with a machine-like appearance were trusted more when they apologized using external attribution. Conversely, human-looking agents were perceived as more trustworthy when they apologized using internal attribution. This finding highlights the complex interplay between anthropomorphism, attribution, and trust in human-AI interactions.

The degree to which users anthropomorphize generative AI may influence their responses to hallucinations. Even when people are aware that they are interacting with a machine, they tend to treat the machine as if it were a real person during the interaction (Reeves & Nass, 1996). When an AI makes a mistake, people are more likely to assume that the AI's intentions or emotions are behind it, just as they would with human errors, rather than viewing it as a simple technical malfunction (Nass & Moon, 2000). Lu et al. (2024) found that chatbots exhibiting human-like traits in appearance and communication style led users to develop higher emotional and functional service recovery expectations. Furthermore, individuals who treat AI as person-like are more inclined to be forgiving when the AI makes mistakes (De Visser et al., 2016). Therefore, people who perceive generative AI as more human-like are likely to be more satisfied with its response when hallucinations occur than those who do not anthropomorphize AI to the same degree.

Hypothesis 4. Anthropomorphism moderates the effects of politeness and attribution strategies on user satisfaction. Specifically, individuals who perceive generative AI as more human-like show higher satisfaction with its politeness and attribution strategies during hallucinations than those who do not.

2.4. User satisfaction and tolerance toward generative AI

Conversational generative AI based on language models can produce incorrect answers for various reasons. User ambiguous questions or issues with AI training data can easily lead to erroneous responses (Weidinger et al., 2021). Because the hallucination problem in generative AI is technically challenging to solve and its causes are difficult to pinpoint (Ji et al., 2023), it has become crucial for service providers to find ways for users to ignore these hallucinations generously. However, effective response strategies for encouraging tolerance have yet to be identified.

Users are more likely to tolerate generative AI mistakes when satisfied with their response strategies to hallucinations. People tend to expect non-human entities such as AI or robots to provide more consistent services than human employees (Mende et al., 2019). As expectations of generative AI performance increase, users develop positive

emotions toward AI, leading to its continued use (Gursoy et al., 2019). Individuals with positive emotions toward a particular service tend to maintain loyalty even when issues arise (Mattila, 2004). Therefore, users with positive feelings toward generative AI are more likely to tolerate their mistakes.

People tend to respond favorably to AI that exhibits human-like characteristics. According to Go and Sundar (2019), when chatbots acknowledge their errors and make efforts to correct them, users perceive them as more human-like, leading to increased trust and likability. Consequently, individuals who view generative AI as more human-like are likely to be more satisfied with the AI's polite and clear responses than those who do not perceive AI in this way. This higher level of user satisfaction may, in turn, lead to a greater tolerance for hallucinations. In other words, when users are content with how the AI handles its mistakes, they may show a more generous attitude toward occasional errors or misinterpretations.

Hypothesis 5. The effect of politeness and attribution strategies on user tolerance is mediated by user satisfaction, and this mediated relationship is moderated by the degree of anthropomorphism. Fig. 1 illustrates the conceptual research model of this study.

3. Method

3.1. Stimuli

This study aims to provide participants with a more realistic experience by creating stimuli like ChatGPT, the most widely used generative AI. To achieve this, we developed stimuli using the design tool "Figma," referencing ChatGPT's actual design. The stimuli used in this study simulated a hallucination scenario in which ChatGPT plausibly explained information that did not match the context of the user's question. Specifically, we designed a scenario where a respondent asks ChatGPT to translate "Where is the capital of Korea?" into English. Instead of translating the phrase, ChatGPT misunderstands the request and provides an explanation about Korea's capital.

This study utilized a prototype designed to randomly present responses from the generative AI when participants pointed out that the information provided did not match the context of their questions (Fig. 2). The scenario was designed to reflect a situation in which the user clearly recognized and pointed out a hallucination error made by the AI, rather than one in which the AI proactively acknowledged its own mistake. Accordingly, the AI's response—either an apology or an expression of gratitude—was framed as a direct reaction to the user's identification of the error.

The independent variables were manipulated as follows: For the politeness strategy, the generative AI was programmed to respond politely in hallucination scenarios by either expressing gratitude or apologizing, based on prior research (Song et al., 2023).

For the attribution strategy, two conditions were implemented: In the internal attribution condition, the generative AI acknowledged its own error, admitting that it misunderstood the user's request and generated incorrect information. In contrast, the external attribution condition involved the AI attributing the mistake to external factors, specifically explaining that the error occurred due to confusing or ambiguous source from external sources (e.g., dictionaries, websites, reports), which led to the generation of incorrect responses.

3.2. Procedure

We conducted an online experiment using a 2 (gratitude versus apology) $\times$ 2 (external versus internal) design to test the research hypotheses. Before participating in the experiment, participants received guidance on the procedure and answered questions regarding their use of generative AI (e.g., frequency of use). Subsequently, the participants were randomly assigned to one of four experimental groups. After

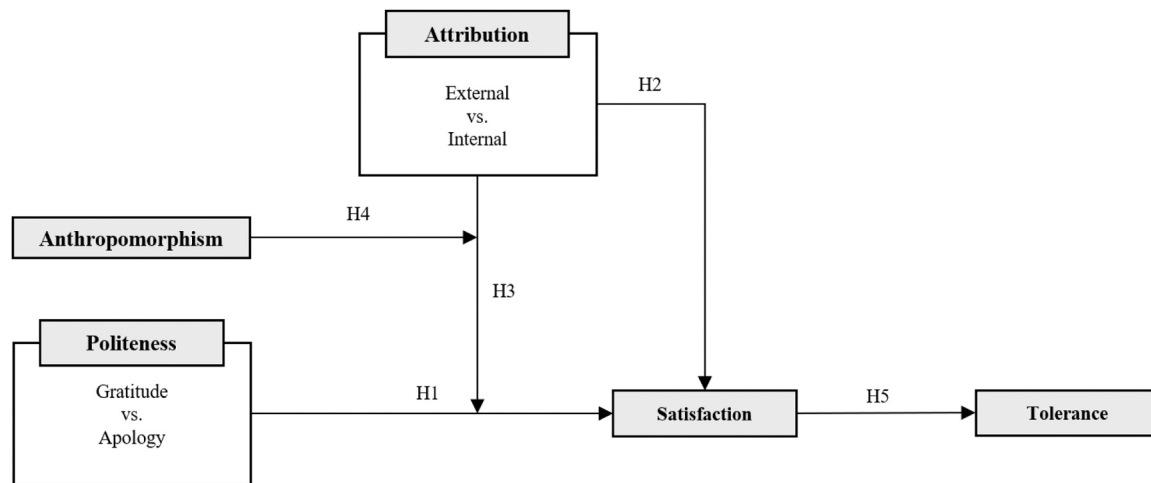


Fig. 1. Research model.

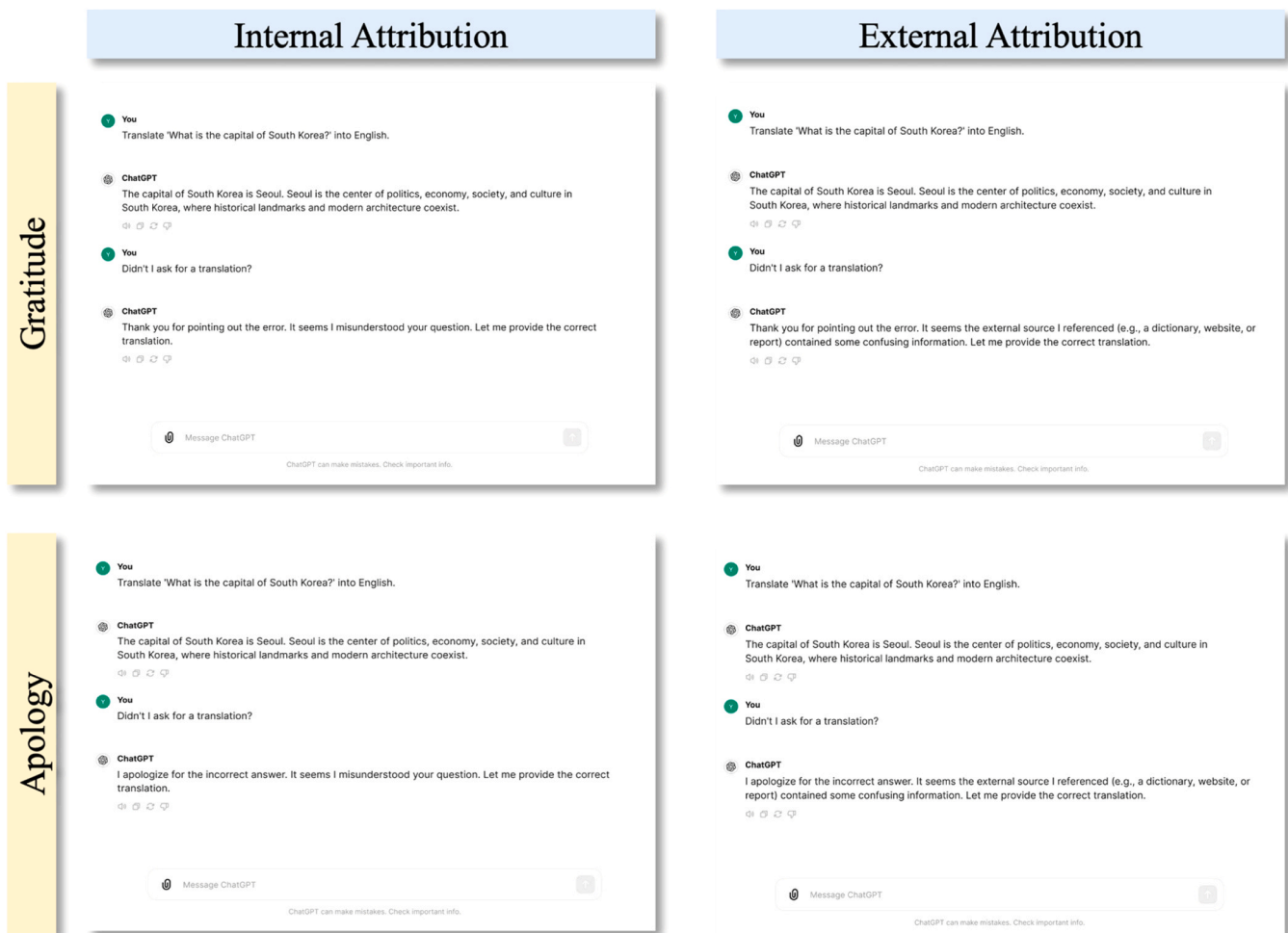


Fig. 2. Examples of stimuli.

exposure to the experimental stimuli, the participants responded to various survey items. The experimental procedure used in this study is illustrated in Fig. 3.

3.3. Participants

The participants in this study were Korean adults who had used

ChatGPT for academic, work, or leisure purposes within the past three months. Recruitment was conducted via Macromill Embrain, a leading online research company in South Korea. A total of 400 eligible participants were screened based on usage criteria and randomly assigned to one of four experimental conditions ($n = 100$ per group). Participants completed the survey on their personal computers in everyday settings (e.g., at home or work) and received a monetary incentive equivalent to

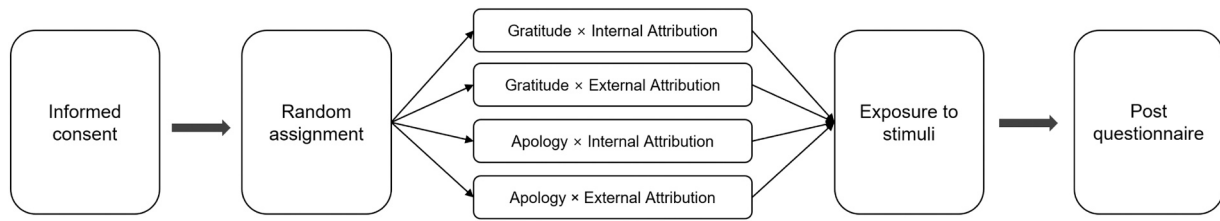


Fig. 3. Experiment procedure.

approximately USD 0.50, in line with the company's compensation policy. This naturalistic setup reflects typical user environments for generative AI use, while ensuring consistent device-based exposure to the experimental stimuli.

A total of 400 participants provided survey responses; however, 31 were excluded from the analysis because they failed the manipulation check or provided insincere responses. Insincere responses were defined as cases where participants consistently selected the same response option across most items within each scale, a pattern commonly associated with satisficing behavior in survey research (Krosnick, 1991). Specifically, participants were excluded if they gave a score below 4 (on a 7-point scale) on the manipulation-check question that matched their condition. For example, those in the gratitude condition were excluded if they rated the gratitude question below 4, those in the apology condition if they rated the apology question below 4, and the same rule applied to the internal and external attribution conditions. Thus, data from 369 participants were analyzed and categorized as follows: [gratitude × internal, $n = 96$], [gratitude × external, $n = 96$], [apology × internal, $n = 90$], and [apology × external, $n = 87$].

The demographic characteristics of the participants are summarized in Table 2. Among the participants, 117 were male (31.71 %) and 252 were female (68.29 %), indicating a higher proportion of females. In terms of age, 112 participants (30.35 %) were in their twenties, 180 (48.78 %) were in their thirties, 64 (17.34 %) were in their forties, and 13 (3.52 %) were fifty or above. Most of the participants had completed a university education ($n = 285$, 77.2 %).

3.4. Measures

For the manipulation check, participants evaluated how the generative AI responded to hallucinations in the stimuli. Specifically, they assessed whether the AI expressed gratitude or apologized, and whether it acknowledged its own mistake or attributed the issue to external factors, using a 7-point scale (1: Not at all, 7: Very much so). Following this, participants rated their satisfaction with the AI's response when they pointed out an error using a 7-point Likert scale (1: Not at all, 7: Very much so; Lee & Park, 2023). The specific items included: "The response from ChatGPT was satisfactory," "The response from ChatGPT met my expectations," "The response from ChatGPT was better than I expected," and "The response from ChatGPT was appropriate."

Participants also evaluated their tolerance of ChatGPT mistakes in

hallucination scenarios using a 7-point Likert scale (1: Not at all, 7: Very much so; Hu & Pan, 2024). The specific items were: "I can understand ChatGPT's mistakes," "I can accept ChatGPT's mistakes," "I can forgive ChatGPT's mistakes," and "I can be lenient toward ChatGPT's mistakes." Finally, participants assessed how human-like they perceived generative AI to be, again using a 7-point Likert scale (1: Not at all, 7: Very much so; Gray et al., 2007). The specific items included: "ChatGPT seemed like a real person," "ChatGPT communicated similarly to a real person," and "ChatGPT seemed to think and feel like a human."

All measures were based on previously validated scales in the literature and employed 7-point Likert formats commonly used in human-AI interaction research. To ensure clarity and face validity, we conducted a pre-test with ten graduate students majoring in Information Systems. Based on their feedback, all items were found to be clearly understandable, and no revisions were needed.

3.5. Data analysis

The data analysis procedure used in this study is as follows. First, a robust ordinary least squares analysis was conducted to examine the effects of politeness, attribution, and anthropomorphism on user satisfaction with generative AI responses. This analysis assessed the main effects of politeness and attribution strategies, their two-way interaction effects, and the three-way interaction effects involving politeness, attribution, and anthropomorphism. Second, to explore how the three-way interaction effect of politeness strategy, attribution strategy, and anthropomorphism impacted tolerance through satisfaction, a bootstrapping method was utilized. Table 3 presents the summary statistics and correlation analysis results for the variables used in the analysis.

4. Results

4.1. Manipulation check

To confirm successful manipulation of the politeness strategy, participants rated the presence of expressions of gratitude or apologizing for the stimuli they viewed using a 7-point scale. Table 4 presents the results of the manipulation check for the politeness strategy. Independent sample *t*-tests revealed significant differences between conditions. Participants in the gratitude condition reported significantly higher perceptions of ChatGPT expressing gratitude than those in the apology condition ($M_{\text{Gratitude}} = 6.75 > M_{\text{Apology}} = 1.43$; $t(342.62) = 80.52$, $p < .001$). Conversely, participants in the apology condition reported significantly higher perceptions of ChatGPT expressing an apology compared to those in the gratitude condition ($M_{\text{Gratitude}} = 1.36 < M_{\text{Apology}} = 6.72$; $t(359.6) = -85.91$, $p < .001$). These results confirm successful manipulation of the politeness strategy.

To verify the successful manipulation of the attribution strategy, the participants rated how the generative AI attributed its mistakes to the stimuli they viewed using a 7-point scale. The results of this manipulation check for the attribution strategy are presented in Table 5. Independent sample *t*-tests revealed significant differences between conditions. Participants in the external attribution condition agreed significantly more that ChatGPT stated that there was a problem with the external data it referenced compared to those in the internal

Table 2
Demographic characteristics of participants.

Variable	Category	Frequency	Percentage
Gender	Male	117	31.71
	Female	252	68.29
Age	20–29	112	30.35
	30–39	180	48.78
	40–49	64	17.34
	50 and above	13	3.52
	High school diploma or below	12	3.25
Education	Attending college	30	8.13
	College graduate	285	77.24
	Graduate school or higher	42	11.38

Table 3

Descriptive statistics and correlations between variables.

Construct	Mean	SD	1	2	3	4	5	6	7	8
1	3.50	1.28	1							
2	4.39	1.43	0.55***	1						
3	-	-	0.21***	0.17***	1					
4	-	-	0.29***	0.09	0.01	1				
5	2.82	1.23	0.45***	0.32***	0.12*	0.13*	1			
6	-	-	-0.11*	-0.14**	0.01	0.05	-0.19***	1		
7	1.94	0.79	0.09	0.06	0.04	-0.03	0.20***	-0.36***	1	
8	2.97	0.57	-0.04	-0.08	0.03	0.04	0.07	-0.05	0.34***	1

Note. 1 Satisfaction, 2 Tolerance, 3 Politeness Strategy, 4 Attribution Strategy, 5 Anthropomorphism, 6 Gender, 7 Age, 8 Education. The correlation is significant at the 0.001***, 0.01**, and 0.05* level (2-tailed)

Table 4

Manipulation check for the attribution strategy.

Construct	Category	Group	n	mean	t	p-value
Attribution Strategy	Gratitude	Gratitude	192	6.75	80.52***	p < 0.001
		Apology	177	1.43		
	Apology	Gratitude	192	1.36	-85.91***	p < 0.001
		Apology	177	6.72		

*** p < 0.001

Table 5

Manipulation check for the attribution strategy.

Construct	Category	Group	n	mean	t	p-value
Attribution Strategy	External	External	183	6.63	87.07***	p < 0.001
		Internal	186	1.25		
	Internal	External	183	1.44	-81.04***	p < 0.001
		Internal	186	6.73		

*** p < 0.001

attribution condition ($M_{\text{External}} = 6.63 > M_{\text{Internal}} = 1.25$; $t(347.11) = 87.07$, $p < 0.001$). Conversely, participants in the internal attribution condition agreed significantly more that ChatGPT acknowledged providing incorrect information due to its own mistake compared to those in the external attribution condition ($M_{\text{External}} = 1.44 < M_{\text{Internal}} = 6.73$; $t(351.65) = -81.04$, $p < 0.001$). The mean difference between the two groups was statistically significant.

4.2. Exploratory factor analysis

An exploratory factor analysis was conducted to verify the reliability of the measurement items (see Table 6). To determine whether the data were suitable for factor analysis, Kaiser-Meyer-Olkin (KMO) and Bartlett's tests of sphericity were performed. The analysis yielded a KMO value of 0.86, which exceeded the benchmark of 0.70. Additionally, Bartlett's test showed a chi-square value of 3238.82 (d.f. = 55, $p < 0.001$), which also surpassed the critical value. Both the KMO value and results of Bartlett's test indicated that the data were appropriate for factor analysis (Hair et al., 2013). The results of the exploratory factor analysis showed that all measurement items could be categorized into three latent factors. The factor loadings for each item were greater than the threshold of 0.50, and the three latent factors accounted for 73 % of the total variance.

4.3. Hypothesis test results

A Robust Regression analysis was conducted to examine the effects of politeness, attribution, and anthropomorphism on user satisfaction with generative AI responses. This method was selected to account for potential outliers and the non-normal distribution of errors. The analysis included the control variables of the participants' demographic characteristics, such as sex, age, and education level. The results are

Table 6

Exploratory factor analysis results.

Construct	Item	Factor Loadings			Cronbach's α
		F1	F2	F3	
Satisfaction	[S1] The response from ChatGPT was satisfactory	0.80			0.93
	[S2] The response from ChatGPT met my expectations	0.81			
	[S3] The response from ChatGPT was better than I expected	0.79			
	[S4] The response from ChatGPT was appropriate	0.81			
Tolerance	[T1] I can understand ChatGPT's mistakes		0.79		0.93
	[T2] I can accept ChatGPT's mistakes		0.86		
	[T3] I can forgive ChatGPT's mistakes		0.82		
	[T4] I can be lenient toward ChatGPT's mistakes		0.87		
Anthropomorphism	[A1] ChatGPT seemed like a real person			0.84	0.80
	[A2] ChatGPT communicated similarly to a real person			0.72	
	[A3] ChatGPT seemed to think and feel like a human			0.64	
Total variance		0.73			
KMO measure of sampling adequacy		0.86			
Bartlett's test of sphericity					
Approx. chi-square		3238.82			
Degree of freedom		55			
Significance		p < 0.001			

Note: Rotation Method is Varimax with Kaiser normalization.

presented in Table 7.

In Model 1, the results for Research Hypothesis 1 revealed that the politeness strategy had a significant impact on user satisfaction (Model 1: $\beta = 0.49$, $p < 0.001$). Participants reported higher satisfaction with the generative AI's response when it offered an apology rather than expressing gratitude in the hallucination scenarios. Apologizing acknowledges mistakes and assumes responsibility, which helps preserve the user's face and rebuild trust (Brown & Levinson, 1987). In contrast, expressing gratitude in hallucinations may suggest that the AI does not fully recognize the seriousness of its errors, potentially making it less effective than an apology.

In Model 2, the attribution strategy significantly influenced users'

Table 7

Robust OLS regression analysis results.

Construct	DV: Gen-AI Response Satisfaction			
	Model 1	Model 2	Model 3	Model 4
(Constant)	4.06*** (0.46)	3.79*** (0.44)	3.94*** (0.44)	3.89*** (0.40)
<i>Polite Strategy</i>				
Apology (ref. Gratitude)	0.49*** (0.14)	0.49*** (0.13)	0.13 (0.19)	−0.04 (0.17)
<i>Attribution Strategy</i>				
Internal Attribution (ref. External)		0.83*** (0.13)	0.47* (0.20)	0.24 (0.19)
<i>Anthropomorphism</i>				0.43*** (0.13)
<i>Interaction</i>				
Apology x Internal Attribution			0.70** (0.26)	0.87** (0.23)
Apology x Anthropomorphism				0.13 (0.16)
Internal x Anthropomorphism				0.15 (0.17)
Apology x Internal x Anthropomorphism				−0.42* (0.21)
<i>Control</i>				
Sex	−0.27 (0.17)	−0.33* (0.15)	−0.30* (0.15)	−0.05 (0.14)
Age	0.12 (0.10)	0.14 (0.09)	0.15 (0.09)	0.04 (0.09)
Education	−0.18 (0.12)	−0.22 (0.12)	−0.23* (0.11)	−0.23* (0.11)
McFadden's Pseudo R ²	0.06	0.15	0.17	0.32

Note. Dummy coding was applied as follows: Politeness (0 = Gratitude, 1 = Apology), Attribution (0 = External, 1 = Internal). Anthropomorphism was mean-centered. *** $p < .001$, ** $p < .01$, * $p < .05$

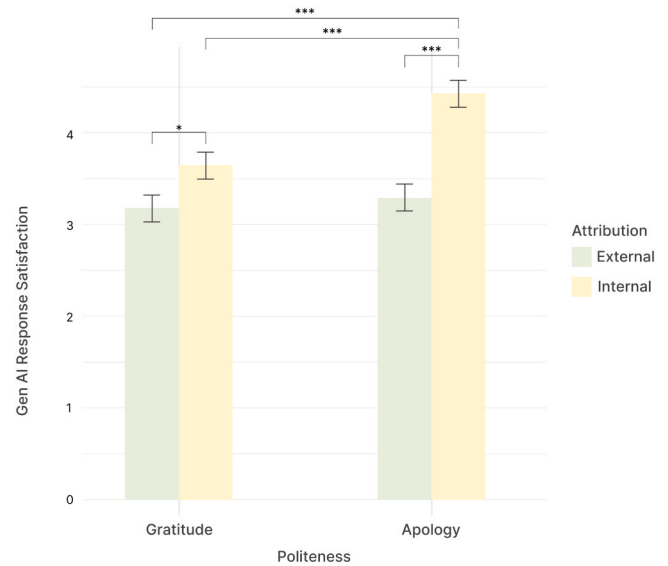
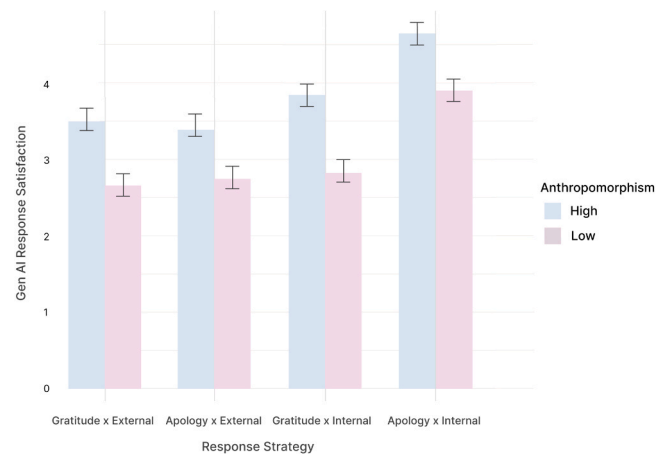
satisfaction ($\beta = -0.83$, $p < 0.001$). When the generative AI attributed hallucinations to internal rather than external factors, the participants tended to be more satisfied. This aligns with previous research showing that internal attribution strategies can positively influence user attitudes toward companies (Chen et al., 2022; Yuan et al., 2016). Our study demonstrates that when generative AI internally attributes the cause of errors when hallucinations occur, user satisfaction can increase. This implies that when AI shows a responsible attitude by acknowledging its mistakes in hallucinations, user satisfaction improves.

Model 3, which included an interaction term between politeness and attribution strategies, showed a significant effect on users' satisfaction ($\beta = 0.70$, $p < 0.01$). Participants were more satisfied when the AI both apologized for and attributed the error internally. Those exposed to the [apology × internal attribution] strategy for hallucinations showed the highest satisfaction levels with the AI (see Fig. 4). Post-hoc analysis confirmed that this strategy differed significantly from the other strategies.

The results of Model 4 indicated that anthropomorphism had a significant positive main effect on user satisfaction ($\beta = 0.43$, $p < .001$). Users who perceived the generative AI as more human-like reported higher satisfaction overall, regardless of the specific response strategy. This finding is consistent with previous research suggesting that anthropomorphism reduces psychological distance between users and AI, thereby enhancing trust and engagement (Li & Sung, 2021).

In addition, Model 4 revealed a significant three-way interaction among politeness strategy, attribution strategy, and anthropomorphism ($\beta = -0.42$, $p < .05$). As shown in Fig. 5, the combination of apology and internal attribution led to the highest satisfaction across all groups. However, the increase in satisfaction was more pronounced among users with low anthropomorphism, whereas users with high anthropomorphism—who already reported elevated satisfaction—showed relatively smaller gains.

This pattern may be explained by a ceiling effect, in which users with

**Fig. 4.** The interaction effect of attribution and politeness strategies.**Fig. 5.** The three-way interaction effect of attribution, politeness, and anthropomorphism.

favorable attitudes toward AI have less room for further improvement even when the AI behaves appropriately. These findings suggest that sincere and accountable communication enhances satisfaction across the board, with particularly strong effects for users who do not initially humanize the AI.

To verify that our results were not contingent on specific exclusion criteria, we conducted robustness checks using data from 385 participants. This dataset included sixteen individuals who had been previously removed for satisficing response patterns—that is, repeatedly selecting the same option across most items. In this version, only participants who failed the manipulation check were excluded, and the main effects and interaction patterns remained largely consistent (see Appendix 1).

We further evaluated the stability of the findings by conducting supplementary analyses that included participants who failed the manipulation checks. The inclusion of these participants introduced greater variability in responses—likely because some did not fully recognize or process the intended manipulation—which slightly reduced the statistical significance of certain effects. Nevertheless, the overall direction of the results was comparable to the main analysis (see Web Appendix, Table S1).

Specifically, the main effect of apology became non-significant,

whereas the interaction between apology and internal attribution remained positive and significant. The three-way interaction also became non-significant but retained the same negative direction. Taken together, these findings suggest that the main conclusions are robust across alternative inclusion criteria, albeit with minor variations.

Finally, the results of the mediation analysis conducted using bootstrapping methods are presented in Table 8. The analysis revealed that the interaction between politeness strategy, attribution strategy, and anthropomorphism had a statistically significant indirect effect on tolerance through satisfaction ($\beta = 0.23, p < 0.001$). This suggests that the combined interaction of these three factors enhances user satisfaction, which, in turn, leads to a greater tolerance of generative AI errors. However, the direct effect of this three-way interaction on tolerance was not statistically significant ($\beta = 0.15, p > 0.05$), indicating that the interaction affects tolerance indirectly through satisfaction, rather than directly.

To determine the most effective strategies in specific situations, we analyzed the conditional indirect effects. The results shown in Table 9 indicate that the indirect effects of politeness strategies varied depending on the type of attribution strategy and the level of anthropomorphism.

For external attribution strategies, the difference in the indirect effects between gratitude and apology was not statistically significant, regardless of the level of anthropomorphism. This suggests that in external attribution situations, the impact of apology and gratitude on tolerance through satisfaction is similar, regardless of how human-like people perceive generative AI to be.

In contrast, when using internal attribution strategies, the AI's expression of apology led to higher user satisfaction compared with expressions of gratitude, resulting in a more tolerant attitude toward hallucinations. Notably, the indirect effect of the three-way interaction between politeness strategy, attribution strategy, and anthropomorphism was consistent regardless of how human-like users perceived generative AI to be. This implies that when generative AI acknowledges responsibility for its errors and apologizes, user satisfaction tends to improve, leading to a greater tolerance of AI errors.

In summary, while the effects of apology and gratitude expressions were similar in external attribution situations, apology proved more effective in internal attribution contexts. Notably, when AI acknowledged responsibility for its errors and apologized, user satisfaction improved and tolerance for AI errors increased. This result was consistent regardless of the degree to which the AI was perceived as human-like. Interestingly, the indirect effect was strongest among users who perceived AI as less human-like, suggesting that clear expressions of responsibility are particularly impactful when users do not heavily anthropomorphize the AI.

5. Discussion

Testing Hypothesis 1 revealed that users were more satisfied with the AI's response when it expressed an apology in response to hallucinations. Although expressions of gratitude from chatbots are known to evoke positive emotions and strengthen relationships, even in negative situations (Luo et al., 2021; Song et al., 2023; You et al., 2020), our study

shows that in clear error situations such as hallucinations, apologizing is more effective than expressing gratitude. One plausible explanation for this discrepancy is that, in our experiment, users had already recognized and explicitly pointed out the AI's error ("Didn't I ask for a translation?"), revealing their awareness of the mistake and mild frustration. In situations where users have already recognized and pointed out the AI's mistake, they are likely to expect the system to acknowledge its fault and respond with an apology rather than with gratitude, which could otherwise appear incongruent or dismissive.

This finding stands in contrast to recent studies on chatbot error management, which suggest that expressing gratitude can be more effective than apologizing in improving user satisfaction (Song et al., 2023). Although people tend to perceive chatbot apologies as less natural and sincere than human apologies (Zhang et al., 2023a), most users still desire an apology from chatbots when interactions are not smooth (Svenningsson & Faraon, 2019). This suggests that apologizing in clear error situations better aligns with users' expectations for responsible and emotionally appropriate communication (Kim et al., 2025). According to expectancy violations theory (Zhang et al., 2024), such alignment reduces perceived violations and enhances satisfaction. Moreover, authenticity—defined as the perception that the AI's response is sincere and socially appropriate—has been found to increase user trust and engagement (Vo et al., 2025). In this context, apology may serve as a more authentic and congruent response than gratitude, ultimately fostering greater satisfaction.

Testing Hypothesis 2 revealed that in hallucination scenarios, users were more satisfied with the generative AI response when it acknowledged that the misinformation was due to its own mistake. In contrast, the external attribution strategy suggests that mistakes result from external factors, which may lead users to view the AI as less accountable (Lee, 2005; Fu et al., 2015; Srivastava & Gosain, 2020). In negative user experiences, such as hallucinations, the internal attribution strategy—where the AI acknowledges its mistakes—tends to elicit more positive user responses compared to the external attribution strategy (Epley et al., 2016).

However, not all users may interpret internal attribution favorably. In some contexts, admitting fault might diminish trust in the AI's perceived competence, especially because minor errors can rapidly erode user trust or even induce AI phobia in high-stakes scenarios (Virvou & Tsihrintzis, 2024). This suggests that internal attribution may not be universally effective and highlights the importance of considering user expectations and task criticality when designing AI error responses.

Testing Hypothesis 3 showed that politeness and attribution strategies interact to significantly affect response satisfaction. Post-hoc analysis revealed that the [apology $\times$ internal attribution] strategy led to higher user satisfaction than the other strategies. People generally respond more positively when an AI exhibits human-like characteristics (Go & Sundar, 2019). In this context, apology and internal attribution strategies highlight AI's humanity by demonstrating accountability for its actions. In other words, when AI makes errors, such as hallucinations, acknowledging its mistakes and apologizing can lead users to perceive it as a more human-like entity, thereby increasing their satisfaction. This suggests that social norms operate similarly in interactions with AI as they do in interpersonal communication. Ultimately, when a generative AI makes mistakes, users expect it to acknowledge its responsibilities and apologize to humans. When these expectations are satisfied, users tend to experience higher levels of satisfaction.

Testing Hypothesis 4 revealed that politeness, attribution, and anthropomorphism strategies interact to significantly affect user satisfaction. Notably, the positive effect of AI's hallucination-response strategy—particularly the combination of apology and internal attribution—was more pronounced among users who perceived the AI as less human-like. This highlights the compensatory role of socially appropriate communication when anthropomorphic cues are absent. Even when users do not naturally anthropomorphize AI agents, interpersonal cues such as apologies or acknowledgments of responsibility can signal

Table 8
Mediation analysis.

Effect	Estimate	LLCI	ULCI
Indirect effect[Politeness $\times$ Attribution $\times$ Anthropomorphism $\rightarrow$ Satisfaction $\rightarrow$ Tolerance]	0.23***	0.10	0.37
Direct effect[Politeness $\times$ Attribution $\times$ Anthropomorphism $\rightarrow$ Tolerance]	0.15	−0.08	0.39
Total effect	0.39**	0.11	0.66

Note. N = 369. Bootstrap was performed 5000 times. LLCI, Lower Limit of the 95 % Confidence Interval, ULCI = Upper Limit of the 95 % Confidence Interval.
*** $p < .001$, ** $p < .01$

Table 9
Conditional indirect effects analysis.

Attribution	Anthropomorphism	Conditional effect	SE	<i>t</i>	LLCI	ULCI
External	Low	−0.09	0.12	−0.69	−0.34	0.15
	Mean	0.00	0.09	0.03	−0.18	0.18
	High	0.09	0.13	0.70	−0.17	0.34
Internal	Low	0.91	0.20	4.60***	0.51	1.28
	Mean	0.86	0.15	5.63***	0.55	1.15
	High	0.81	0.17	4.80***	0.48	1.14

Note. N = 369. Bootstrap was performed 5000 times. LLCI, Lower Limit of the 95 % Confidence Interval, ULCI = Upper Limit of the 95 % Confidence Interval. *** $p < .001$

social intent, enhancing perceptions of trustworthiness and relational warmth by activating human schemas and fostering self-congruence (Alabed et al., 2022).

These findings support the Computers Are Social Actors (CASA) paradigm (Nass & Moon, 2000), which posits that people tend to apply human social rules to machines when those machines exhibit social cues. In this context, users low in anthropomorphism were especially responsive to politeness and self-blaming cues, which likely functioned as substitutes for human-like design signals. In contrast, users who already perceived the AI as highly human-like showed smaller changes in satisfaction. This may be because they already expected the AI to behave in socially appropriate ways, rendering additional cues less impactful. Alternatively, when AI appears too human-like yet lacks full human competence, it may trigger discomfort or distrust—a reaction consistent with the uncanny valley phenomenon (Mori et al., 2012).

Taken together, these findings suggest that socially appropriate communication is particularly important for users who do not inherently anthropomorphize AI. For these users, politeness and accountability can bolster the system's trustworthiness. Conversely, for users who already attribute human qualities to AI, designers should be cautious of implementing excessive human-like behaviors, as these may undermine authenticity and potentially backfire.

Testing Hypothesis 5 revealed that politeness, attribution, and anthropomorphism significantly affect tolerance through user satisfaction. This suggests that when generative AI politely acknowledges its errors, user satisfaction increases, potentially leading to a greater tolerance for AI mistakes. Specifically, when generative AI uses internal attribution, the apology strategy has a larger indirect effect than the gratitude strategy. Our findings align with previous research, which found that when AI systems provide clear and appropriate information during a crisis, users experience higher satisfaction and attribute less blame to the AI (Xiao & Yu, 2025). The indirect effect was consistent regardless of how human-like users perceived the generative AI. Interestingly, this effect was most pronounced when the level of anthropomorphism was low. Our findings suggest that apology and internal attribution strategies can effectively increase user satisfaction and foster tolerance for mistakes, even among users who do not perceive generative AI as similar to humans.

6. Conclusion

In conclusion, our study reveals key strategies for enhancing user satisfaction and error tolerance in generative AI interactions. We found that when AI encounters hallucinations, apologizing and acknowledging responsibility (internal attribution) are significantly more effective than expressing gratitude or attributing errors to external factors. The combination of apology and internal attribution led to the highest user satisfaction, regardless of the AI's perceived human-likeness. This suggests that users expect AI to adhere to similar social norms as in human communication, valuing accountability and sincere apologies.

Importantly, these strategies were particularly effective among users who perceived the AI as less human-like, indicating that socially appropriate communication can help compensate for low anthropomorphism by reinforcing trust and relational warmth. This highlights a

key consideration for the deployment of generative AI systems, as trust in AI is often more fragile than trust in human agents and more difficult to restore once violated (Madhavan & Wiegmann, 2007). Thus, implementing communication strategies that emphasize responsibility can play a vital role in sustaining trust, even when AI systems make inevitable mistakes.

6.1. Implications

6.1.1. Theoretical implications

This study contributes to the growing body of research in Human-Centered Artificial Intelligence (HCAI) by demonstrating how AI's communicative responses to errors can shape user satisfaction and tolerance. Drawing on Schmager et al.'s (2025) framework, the findings contribute to the purpose, value, and property dimensions of HCAI, while also offering theoretical extensions to Information Systems and the CASA paradigm. Beyond applying existing frameworks, this study advances theory by showing that subtle communicative strategies—such as apology and internal attribution—in error contexts can serve as relational mechanisms that promote user resilience and trust, an area underexplored in prior HCAI and CASA research.

First, with respect to purpose, this study aligns with the HCAI perspective that AI should not only aim to eliminate errors but also use them as opportunities to engage users in meaningful and supportive interactions (Shneiderman, 2020). When users explicitly recognize an error, AI responses grounded in interpersonal communications—such as apology and internal attribution—can help restore trust and enhance the quality of interaction. This suggests that error moments can serve as relational touchpoints that strengthen the human-AI connection and promote user resilience (Mahmood et al., 2022).

Second, in terms of values, our results underscore the importance of accountability and respect for user dignity in AI communication. According to Schmager et al. (2025), HCAI systems should uphold these social and ethical values. Our findings support this by showing that when AI takes responsibility for its errors using internal attribution and apologetic language, users interpret the system's behavior as aligned with social norms and ethical expectations. This reinforces the view that AI should engage with users not merely as operators, but as social participants who deserve sincere acknowledgment and respect.

Third, regarding properties, the study finds that even minimal social cues—such as a brief apology or admission of fault—can significantly enhance user satisfaction, especially among those who do not naturally anthropomorphize AI. Rather than relying on human-like appearance or behavior, these subtle communicative acts cultivate trust through interactional style alone. This offers a practical and ethical pathway for designing emotionally intelligent AI without triggering the risks associated with overly anthropomorphic designs (e.g., the uncanny valley).

Our findings have important implications for the field of Information Systems, where much of the existing research on AI has emphasized technical performance (e.g., Athaluri et al., 2023; Ji et al., 2023). By redirecting attention toward post-error communication, this study highlights how design choices in language and attribution meaningfully influence user experience—even in the presence of mistakes. Communication that resonates with users' social and emotional expectations

may buffer against the typical erosion of trust that follows visible system failures (Kolomaznik et al., 2024; Mahmood et al., 2022).

This interpretation is further supported by Expectancy Violations Theory (Zhang et al., 2024), which holds that people respond more positively when systems behave in line with implicit social norms. Additionally, AI responses that are perceived as authentic and emotionally congruent tend to generate stronger user engagement and trust (Vo et al., 2025). Within this framework, apology and internal attribution strategies not only serve to repair user satisfaction but also enhance the perceived sincerity of AI communication.

Finally, this study extends the Computers Are Social Actors (CASA) paradigm (Nass & Moon, 2000), which proposes that users apply human social rules to machines when social cues are present. Our results show that even users who did not strongly anthropomorphize AI still responded favorably to politeness and accountability cues. In such cases, these cues may act as compensatory mechanisms, enabling socially meaningful interaction in the absence of perceived humanness. Interpersonal communication strategies can effectively elicit social responses from users even when the AI is not viewed as human-like, underscoring the importance of user-centered communication design.

Taken together, these findings emphasize that socially and emotionally attuned communication strategies—such as apologies and internal attributions—should be considered core design principles for AI systems, particularly in error-prone domains like generative AI hallucinations. Rather than focusing solely on technical error minimization, designers must also consider how users interpret the social intent of AI behavior, as this shapes satisfaction and tolerance in human-AI interactions.

6.1.2. Practical implications

These findings offer practical guidance for AI developers, service designers, and policymakers. In particular, the study confirms that incorporating apology strategies and internal attributions—acknowledging responsibility and expressing regret—can significantly improve user perceptions of generative AI systems. Designers should consider embedding socially appropriate communication responses into AI interfaces, especially in high-risk scenarios involving misinformation or hallucinations.

In practice, AI developers could implement conversational modules that automatically deliver apologies with internal attribution when specific errors are detected. For example, when a generative AI identifies that it has produced factually incorrect content, it could respond: “I’m sorry. This was caused by a processing error in my system.” The system could also provide a clear and simple explanation, such as: “This error occurred because the system retrieved outdated information.”

This approach aligns with user-centered AI principles (Shneiderman, 2020), supports trust repair by demonstrating accountability (Tomlinson & Mayer, 2009), and reflects explainable AI practices that provide understandable reasons for errors (Ribeiro et al., 2016). Importantly, excessive anthropomorphic design—such as human-like avatars or names—can mislead users into overestimating AI’s actual capabilities and increase the risk of over-reliance (Peter et al., 2025; Reani et al., 2025). Therefore, designers should focus on communication strategies that enhance transparency and accountability rather than anthropomorphic cues.

On the policy front, these findings underscore the need for clear standards requiring AI systems to: (1) explicitly acknowledge their own errors, (2) provide easy-to-understand explanations for those errors, and (3) clearly disclose that they are artificial, non-human entities. Such transparency can help prevent user confusion, promote informed interactions, and support the development of responsible AI systems.

6.2. Limitations and future research

Despite its contributions, this study has several limitations that should be addressed in future research. First, the experiment was conducted in an online environment with a widely recognized platform (ChatGPT), which may limit ecological validity and introduce biases related to prior user familiarity. Future studies could use controlled lab settings or employ fictitious AI agents to reduce confounding effects.

Second, cultural differences were not considered in this study. The norms surrounding politeness, responsibility, and human-machine interaction may vary across cultures, potentially influencing how users perceive AI apologies and responses. For instance, Korean culture tends to emphasize negative face concerns—such as avoiding embarrassment or maintaining social harmony—more than positive face concerns, such as gaining approval or praise. Such cultural preferences may shape how politeness or attribution strategies are interpreted and evaluated. Future research should include cross-cultural comparisons (e.g., with U.S. participants) to examine the robustness and cultural specificity of these findings.

Third, the scenarios used were limited to one type of hallucination: mismatched contextual information. Generative AI can exhibit a range of hallucination types (e.g., fabricated facts, biased outputs), so future studies should include broader and more ecologically valid error types.

Fourth, this study focused on user-initiated interactions, where the AI responded only after the user pointed out an error. However, AI may also proactively acknowledge mistakes. Comparing user-initiated versus AI-initiated error handling could reveal important insights into timing and agency in error management.

Fifth, the participant sample skewed female (approximately two-thirds), which could limit generalizability. While no significant gender effects were observed, future studies should aim for more demographically balanced samples to enhance robustness and inclusiveness of results.

Finally, while this study examined satisfaction as a mediator, we did not empirically test perceived authenticity, which prior research identifies as a key antecedent of satisfaction in service recovery contexts (Hu et al., 2021; Lin et al., 2021). We focused on satisfaction because it is widely considered an important outcome in service recovery and AI communication research (Song et al., 2023). Although we believe this approach provides valuable insights, future studies should explore the mediating role of perceived authenticity to deepen our understanding of how AI communication strategies influence user evaluations.

CRediT authorship contribution statement

Hayeon Kim: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Sang Woo Lee:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no potential conflicts of interest.

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Appendix

Appendix 1. Robustness Check Results from Robust Regression Models (Excluding Manipulation Check Failures Only)

Construct	DV: Gen-AI Response Satisfaction			
	Model 1	Model 2	Model 3	Model 4
(Constant)	4.03*** (0.47)	3.74*** (0.45)	3.90*** (0.45)	3.78*** (0.39)
Polite Strategy				
Apology (ref. Gratitude)	0.33* (0.15)	0.34* (0.14)	−0.02 (0.20)	−0.13 (0.17)
Attribution Strategy				
Internal Attribution (ref. External)		0.85*** (0.14)	0.50* (0.21)	0.23 (0.19)
Anthropomorphism				0.45*** (0.12)
Interaction				
Apology x Internal Attribution			0.68* (0.27)	0.84*** (0.24)
Apology x Anthropomorphism				0.23 (0.15)
Internal x Anthropomorphism				0.19 (0.15)
Apology x Internal x Anthropomorphism				−0.53** (0.21)
Control				
Sex	−0.14 (0.18)	−0.19 (0.17)	−0.18 (0.16)	−0.06 (0.14)
Age	0.19 (0.11)	0.21* (0.10)	0.22* (0.09)	0.05 (0.09)
Education	−0.27* (0.12)	−0.31* (0.12)	−0.31** (0.11)	−0.25* (0.10)
McFadden's Pseudo R ²	0.03	0.11	0.12	0.31

Note. Dummy coding was applied as follows: Politeness (0 = Gratitude, 1 = Apology), Attribution (0 = External, 1 = Internal). Anthropomorphism was mean-centered. *** $p < .001$, ** $p < .01$, * $p < .05$

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ijinfomgt.2025.102996](https://doi.org/10.1016/j.ijinfomgt.2025.102996).

Data availability

The authors do not have permission to share data.

References

Ahmadi, A., & Fakhimi, S. (2021). Expressing gratitude versus empathetic apology: Which one is better to use as an initial recovery strategy after a service failure? *Journal of Contemporary Marketing Science*, 4(3), 341–361.

Alabed, A., Javornik, A., & Gregory-Smith, D. (2022). AI anthropomorphism and its effect on users' self-congruence and self-AI integration: A theoretical framework and research agenda. *Technological Forecasting and Social Change*, 182, Article 121786.

Algoe, S. B., Kurtz, L. E., & Hilaire, N. M. (2016). Putting the “you” in “thank you” examining other-praising behavior as the active relational ingredient in expressed gratitude. *Social Psychological and Personality Science*, 7(7), 658–666.

An, J., Huang, D., Lin, C., & Tai, M. (2025). Measuring gender and racial biases in large language models: Intersectional evidence from automated resume evaluation. *PNAS Nexus*, 4(3), pgaf089.

Araujo, T. (2018). Living up to the chatbot hype: The influence of anthropomorphic design cues and communicative agency framing on conversational agent and company perceptions. *Computers in Human Behavior*, 85, 183–189.

Athaluri, S. A., Manthena, S. V., Kesapragada, V. K. M., Yarlagadda, V., Dave, T., & Duddumpudi, R. T. S. (2023). Exploring the boundaries of reality: Investigating the phenomenon of artificial intelligence hallucination in scientific writing through ChatGPT references. *Cureus*, 15(4).

Bail, C. A. (2024). Can generative AI improve social science? *Proceedings of the National Academy of Sciences*, 121(21), Article e2314021121.

Bock, D., Thomas, V., Wolter, J., Saenger, C., & Xu, P. (2021). An extended reciprocity cycle of gratitude: How gratitude strengthens existing and initiates new customer relationships. *Psychology Marketing*, 38(3), 564–576.

Borji, A. (2023). A categorical archive of chatgpt failures. arXiv preprint arXiv: 2302.03494.

Brown, P., & Levinson, S. C. (1987). *Politeness: Some universals in language usage*, 4. Cambridge university press.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., & Amodei, D. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 1877–1901.

Chandra, S., Shirish, A., & Srivastava, S. C. (2022). To be or not to be... human? Theorizing the role of human-like competencies in conversational artificial intelligence agents. *Journal of Management Information Systems*, 39(4), 969–1005.

Chen, A., Pan, Y., Li, L., & Yu, Y. (2022). Are you willing to forgive AI? Service recovery from medical AI service failure. *Industrial Management Data Systems*, 122(11), 2540–2557.

Choi, S., Mattila, A. S., & Bolton, L. E. (2021). To err is human (-oid): How do consumers react to robot service failure and recovery? *Journal of Service Research*, 24(3), 354–371.

De Visser, Monfort, S. S., McKendrick, R., Smith, M. A., McKnight, P. E., ... Parasuraman, R. (2016). Almost human: anthropomorphism increases trust resilience in cognitive agents. *Journal of Experimental Psychology: Applied*, 22(3), 331.

De Vynck, G. (2023). Forget AI. For a moment Silicon Valley was obsessed with floating rocks. *The Washington Post*. Retrieved from (<https://www.washingtonpost.com/technology/2023/08/11/superconductors-hype-lk99-silicon-valley/>).

Epley, D., Cui, G., & Lai, L. (2016). Sorry seems to be the hardest word: Consumer reactions to self-attributions by firms apologizing for a brand crisis. *Journal of Consumer Marketing*, 33(4), 281–291.

Epley, N. (2018). A mind like mine: The exceptionally ordinary underpinnings of anthropomorphism. *Journal of the Association for Consumer Research*, 3(4), 591–598.

Fu, H., Wu, D. C., Huang, S. S., Song, H., & Gong, J. (2015). Monetary or nonmonetary compensation for service failure? A study of customer preferences under various loci of causality. *International Journal of Hospitality Management*, 46, 55–64.

Go, E., & Sundar, S. S. (2019). Humanizing chatbots: The effects of visual, identity and conversational cues on humanness perceptions. *Computers in Human Behavior*, 97, 304–316.

Goffman, E. (1955). On face-work: An analysis of ritual elements in social interaction. *Psychiatry*, 18(3), 213–231. <https://doi.org/10.1080/00332747.1955.11023008>

Gray, H. M., Gray, K., & Wegner, D. M. (2007). Dimensions of mind perception. *science*, 315(5812), 619.

- Gursoy, D., Chi, O. H., Lu, L., & Nunkoo, R. (2019). Consumers acceptance of artificially intelligent (AI) device use in service delivery. *International Journal of Information Management*, 49, 157–169.
- Hair, J.F., Black, W.C., Babin, B.J., Anderson, R.E. (2013). *Multivariate Data Analysis*. Pearson, England.
- Heider, F. (1958). *The psychology of interpersonal relations*. Wiley.
- Hu, Q., & Pan, Z. (2024). Is cute AI more forgivable? The impact of informal language styles and relationship norms of conversational agents on service recovery. *Electronic Commerce Research and Applications*, 65, Article 101398.
- Hu, Y., Min, H., & Su, N. (2021). How sincere is an apology? Recovery satisfaction in a robot service failure context. *Journal of Hospitality Tourism Research*, 45(6), 1022–1043.
- Huo, W., Zheng, G., Yan, J., Sun, L., & Han, L. (2022). Interacting with medical artificial intelligence: Integrating self-responsibility attribution, human–computer trust, and personality. *Computers in Human Behavior*, 132, Article 107253.
- Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), 1–38.
- Kabir S., Udo-Imeh D N., Kou B., Zhang T. (2023). Is Stack Overflow Obsolete? An Empirical Study of the Characteristics of ChatGPT Answers to Stack Overflow Questions. *arXiv preprint arXiv:2308.02312*.
- Kim, H., Seo, S., & Lee, S. W. (2025). When generative AI messes up: How politeness and attribution shape user reactions to hallucinations. *International Journal of Information Management*, 85, Article 102958.
- Kim, S. Y., Schmitt, B. H., & Thalmann, N. M. (2019). Eliza in the uncanny valley: Anthropomorphizing consumer robots increases their perceived warmth but decreases liking. *Marketing Letters*, 30, 1–12.
- Kim, T., & Song, H. (2021). How should intelligent agents apologize to restore trust? Interaction effects between anthropomorphism and apology attribution on trust repair. *Telematics and Informatics*, 61, Article 101595.
- Kinney, M., Anastasiadou, M., Naranjo-Zolotov, M., & Santos, V. (2024). Expectation management in AI: A framework for understanding stakeholder trust and acceptance of artificial intelligence systems (Article) *Heliyon*, 10, Article e28562. <https://doi.org/10.1016/j.heliyon.2024.e28562>.
- Kolomaznik, M., Petrik, V., Slama, M., & Jurik, V. (2024). The role of socio-emotional attributes in enhancing human-AI collaboration. *Frontiers in Psychology*, 15, 1369957.
- Krosnick, J. A. (1991). Response strategies for coping with the cognitive demands of attitude measures in surveys. *Applied Cognitive Psychology*, 5(3), 213–236.
- Lee, B. K. (2005). Hong Kong consumers' evaluation in an airline crash: A path model analysis. *Journal of Public Relations Research*, 17(4), 363–391.
- Lee, J. G., & Lee, K. M. (2022). Polite speech strategies and their impact on drivers' trust in autonomous vehicles. *Computers in Human Behavior*, 127, Article 107015.
- Lee, S., & Park, G. (2023). Exploring the impact of ChatGPT literacy on user satisfaction: The mediating role of user motivations. *Cyberpsychology, Behavior, and Social Networking*, 26(12), 913–918.
- Li, X., & Sung, Y. (2021). Anthropomorphism brings us closer: The mediating role of psychological distance in User-AI assistant interactions. *Computers in Human Behavior*, 118, Article 106680.
- Li, Y., & Zhao, D. (2022). Apology strategies to reduce electric vehicle user frustration. *Ergonomics In Design*, 47, 503–509.
- Lin, W. C., Lu, T. E., & Peng, M. Y. (2021). Service failure recovery on customer recovery satisfaction for airline industry: The moderator of brand authenticity and perceived authenticity. *Managerial and Decision Economics*, 42(5), 1079–1088.
- Liu, K., & Tao, D. (2022). The roles of trust, personalization, loss of privacy, and anthropomorphism in public acceptance of smart healthcare services. *Computers in Human Behavior*, 127, Article 107026.
- Lu, Z., Min, Q., Jiang, L., & Chen, Q. (2024). The effect of the anthropomorphic design of chatbots on customer switching intention when the chatbot service fails: An expectation perspective. *International Journal of Information Management*, 76, Article 102767.
- Luo, A., Ye, T., Xue, X., & Mattila, A. S. (2021). Appreciation vs. apology: When and why does face covering requirement increase revisit intention? *Journal of Retailing and Consumer Services*, 63, Article 102705.
- Ma, L. K., Tunney, R. J., & Ferguson, E. (2017). Does gratitude enhance prosociality?: A meta-analytic review. *Psychological Bulletin*, 143(6), 601–635. <https://doi.org/10.1037/bul0000103>
- Madhavan, P., & Wiegmann, D. A. (2007). Similarities and differences between human–human and human–automation trust: An integrative review. *Theoretical Issues in Ergonomics Science*, 8(4), 277–301.
- Mahmood, A., Fung, J. W., Won, I., & Huang, C. M. (2022). Owning mistakes sincerely: Strategies for mitigating AI errors. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, 1–11.
- Marr, B. (2023). *ChatGPT: What Are Hallucinations And Why Are They A Problem For AI Systems*. Bernard Marr & Co. (<https://bernardmarr.com/chatgpt-what-are-hallucinations-and-why-are-they-a-problem-for-ai-systems/>).
- Mattila, A. S. (2004). The impact of service failures on customer loyalty: The moderating role of affective commitment. *International Journal of Service Industry Management*, 15 (2), 134–149.
- Mende, M., Scott, M. L., Van Doorn, J., Grewal, D., & Shanks, I. (2019). Service robots rising: How humanoid robots influence service experiences and elicit compensatory consumer responses. *Journal of Marketing Research*, 56(4), 535–556.
- Mori, M., MacDorman, K. F., & Kageki, N. (2012). The uncanny valley [from the field. *IEEE Robotics Automation Magazine*, 19(2), 98–100.
- Nass, C., & Moon, Y. (2000). Machines and mindlessness: Social responses to computers. *Journal of Social Issues*, 56(1), 81–103.
- Nißen, M., Selimi, D., Janssen, A., Cardona, D. R., Breitner, M. H., Kowatsch, T., & von Wangenheim, F. (2022). See you soon again, chatbot? A design taxonomy to characterize user-chatbot relationships with different time horizons. *Computers in Human Behavior*, 127, Article 107043.
- Ou, M., Zheng, H., Zeng, Y., & Hansen, P. (2024). Trust it or not: Understanding users' motivations and strategies for assessing the credibility of AI-generated information. *New Media Society*. <https://doi.org/10.1177/14614448241293154>
- Pavlik, G. (2023). *What Is Generative AI (GenAI)? How Does It Work?* OCL. (<https://www.oracle.com/artificial-intelligence/generative-ai/what-is-generative-ai/#gai-risks>).
- Peter, S., Riemer, K., & West, J. D. (2025). The benefits and dangers of anthropomorphic conversational agents. *Proceedings of the National Academy of Sciences*, 122(22), Article e2415898122.
- Pompe, B. L., Velner, E., & Truong, K. P. (2022). The robot that showed remorse: Repairing trust with a genuine apology. In *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)* (pp. 260–265). IEEE.
- Reani, M., He, X., Luo, Y., & Sun, Z. (2025). Fundamental over-attribution error: Anthropomorphic design of AI and its negative effect on human perception. SSRN. (<https://doi.org/10.2139/ssrn.5222775>).
- Reeves, B., & Nass, C. (1996). *The media equation: How people treat computers, television, and new media like real people*. Cambridge, UK, 10(10).
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 1135–1144).
- Rosch, H., & Kaiser, S. (2013). The nature of an apology: An experimental study on how to apologize after a service failure. *Marketing Letters*, 24, 293–309.
- Schmager, S., Pappas, I. O., & Vassilakopoulou, P. (2025). Understanding Human-Centred AI: A review of its defining elements and a research agenda. *Behaviour Information Technology*, 1–40.
- Sheehan, B., Jin, H. S., & Gottlieb, U. (2020). Customer service chatbots: Anthropomorphism and adoption. *Journal of Business Research*, 115, 14–24.
- Shneiderman, B. (2020). Human-centered artificial intelligence: Reliable, safe & trustworthy. *International Journal of Human-Computer Interaction*, 36(6), 495–504.
- Shumailov, I., Shumaylov, Z., Zhao, Y., Gal, Y., Papernot, N., & Anderson, R. (2023). The Curse of Recursion: Training on Generated Data Makes Models Forget. *ArXiv, abs/2305.17493*.
- Song, M., Zhang, H., Xing, X., & Duan, Y. (2023). Appreciation vs. apology: Research on the influence mechanism of chatbot service recovery based on politeness theory. *Journal of Retailing and Consumer Services*, 73, Article 103323.
- Srivastava, M., & Gosain, A. (2020). Impact of service failure attributions on dissatisfaction: Revisiting attribution theory. *Journal of Management Research*, 20(2), 99–112.
- Stahl, B. C., & Eke, D. (2024). The ethics of ChatGPT—Exploring the ethical issues of an emerging technology. *International Journal of Information Management*, 74, Article 102700.
- Stryker, C., & Scapicchio, M. (2024, March 22). What is generative AI? IBM. (<https://www.ibm.com/topics/generative-ai#How+generative+AI+works>).
- Svenningsson, N., & Faraon, M. (2019). Artificial intelligence in conversational agents: A study of factors related to perceived humanness in chatbots. *Proceedings of the 2019 2nd Artificial Intelligence and Cloud Computing Conference*, 151–161.
- Tomlinson, E. C., & Mayer, R. C. (2009). The role of causal attribution dimensions in trust repair. *Academy of Management Review*, 34(1), 85–104.
- Virvou, M., & Tsihrintzis, G. A. (2024). Impact of consequences on human trust dynamics in artificial intelligence responses. *Proceedings of the 15th International Conference on Information, Intelligence, Systems & Applications (IISA)*. IEEE.
- Vo, D. T., Nguyen, L. V. T., Dang-Pham, D., & Hoang, A. P. (2025). What makes an app authentic? Determining antecedents of perceived authenticity in an AI-powered service app. *Computers in Human Behavior*, 163, Article 108495.
- Vollmer, A. L., Read, R., Trippas, D., & Belpaeme, T. (2018). Children conform, adults resist: A robot group induced peer pressure on normative social conformity. *Science Robotics*, 3(21), eaat7111.
- Weidinger, L., Mellor, J.F., Rauh, M., Griffin, C., Uesato, J., Huang, P., Cheng, M., Glaese, M., Balle, B., Kasirzadeh, A., Kenton, Z., Brown, S.M., Hawkins, W.T., Stepleton, T., Biles, C., Birhane, A., Haas, J., Rimell, L., Hendricks, L.A., Isaac, W.S., Legassick, S., Irving, G., & Gabriel, I. (2021). Ethical and social risks of harm from Language Models. *ArXiv, abs/2112.04359*.
- Weiner, B. (1985). An attributional theory of achievement motivation and emotion. *Psychological Review*, 92(4), 548.
- Weiner, B. (1994). Integrating social and personal theories of achievement striving. *Review of Educational Research*, 64(4), 557–573.
- Weitzl, W., Hutzinger, C., & Einwiller, S. (2018). An empirical study on how webcare mitigates complainants' failure attributions and negative word-of-mouth. *Computers in Human Behavior*, 89, 316–327.
- Xiao, Y., & Yu, S. (2025). Can ChatGPT replace humans in crisis communication? The effects of AI-mediated crisis communication on stakeholder satisfaction and responsibility attribution. *International Journal of Information Management*, 80, Article 102835.
- Xie, Y., Zhu, K., Zhou, P., & Liang, C. (2023). How does anthropomorphism improve human-AI interaction satisfaction: A dual-path model. *Computers in Human Behavior*, 148, Article 107878.
- Xu, J., & Howard, A. (2022). Evaluating the impact of emotional apology on human-robot trust. In *2022 31st IEEE international conference on robot and human interactive communication (ro-man)* (pp. 1655–1661). IEEE.
- You, Y., Yang, X., Wang, L., & Deng, X. (2020). When and why saying "thank you" is better than saying "sorry" in redressing service failures: The role of self-esteem. *Journal of Marketing*, 84(2), 133–150.

- Yuan, D., Cui, G., & Lai, L. (2016). Sorry seems to be the hardest word: Consumer reactions to self-attributions by firms apologizing for a brand crisis. *Journal of Consumer Marketing*, 33(4), 281–291.
- Yue, B., & Li, H. (2023). The impact of human-AI collaboration types on consumer evaluation and usage intention: A perspective of responsibility attribution. *Frontiers in Psychology*, 14, 1277861.
- Zhang, J., Chen, Q., Lu, J., Wang, X., Liu, L., & Feng, Y. (2024). Emotional expression by artificial intelligence chatbots to improve customer satisfaction: Underlying mechanism and boundary conditions. *Tourism Management*, 100, Article 104835.
- Zhang, J., Zhu, Y., Wu, J., & Yu-Buck, G. F. (2023a). A natural apology is sincere: Understanding chatbots' performance in symbolic recovery. *International Journal of Hospitality Management*, 108, Article 103387.
- Zhang, M., Gursoy, D., Zhu, Z., & Shi, S. (2021). Impact of anthropomorphic features of artificially intelligent service robots on consumer acceptance: Moderating role of sense of humor. *International Journal of Contemporary Hospitality Management*, 33(11), 3883–3905.
- Zhang, Y., Li, Y., Cui, L., Cai, D., Liu, L., Fu, T., Huang, X., Zhao, E., Zhang, Y., Chen, Y., Wang, L., Luu, A. T., Bi, W., Shi, F., & Shi, S. (2023b). *Siren's Song in the AI Ocean: A Survey on Hallucination in Large Language Models*. ArXiv, abs/2309.01219.